

GRAINGUARD · REQUIREMENTS

GrainGuard requirements

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DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept
0.1	2026-09-24	Amish Chadha	Initial scaffold

GrainGuard requirements

These are first-pass requirements for the concept. Targets are proposals for review, not yet validated with farmers, and will be checked by calculation at TRL 3 and revised after co-design visits (see GGD-PRB-001). Three requirements are not met by the current concept (R8, R10 roof capacity, R13), and two are at risk (R6 coverage, R9 fumigation); Table 2 gives the status of each.

The **reference case** is a 5.49 m (18 ft) diameter corrugated steel bin with a 5.6 m eave and a full perforated floor, filled with about 95 t (3,760 bu) of shelled corn stored at a target of 15.0 % moisture, wet basis, with one existing aeration fan giving about 0.36 m³/s (760 cfm, 0.2 cfm/bu).

Table 1. Requirements.

ID	Requirement	Target	Verification (TRL 3 or later)
R1	Grain temperature profile	6 or more levels on one center cable, 1.0 m or less apart; ± 0.3 °C; one reading every 10 min	Sensor datasheet; geometry check in the model
R2	Grain moisture estimate from interstitial air	Within ± 0.8 percentage points, wet basis, for interstitial RH of 20 % to 75 % and grain at 0 °C to 30 °C, from readings taken 6 h or more after the fan stops	EMC error analysis with sensor tolerance; later comparison with oven or meter samples
R3	Ambient air measurement	± 0.3 °C and ± 2 % RH inside a radiation shield; sampled every 5 min; decisions on 30 min averages	Datasheet and shield error estimate
R4	Fan decision rule	Automatic mode runs the fan only when the predicted plenum-air EMC is at or below the target moisture (hold mode), or when plenum air is 5 °C or more below the mean grain temperature and its EMC is no more than 1.0 point above target (cool mode); minimum run 30 min, minimum off 15 min, 4 starts per hour or fewer	Logic review against extension guidance; later simulation over a season of weather data
R5	Fail-safe control	In automatic mode the fan stops within 60 s of controller power loss or when sensor data are more than 30 min old; the hand position of the hand-off-auto switch runs the fan with GrainGuard removed; fan running is confirmed by current within 30 s and a mismatch raises an alarm	Circuit review; later bench test
R6	Heating alert	Alarm at the farmhouse within 15 min when any pod rises 3 °C or more above its 7-day minimum with the fan off, or sits 5 °C or more above the bin median	Logic review; later heated-core test
R7	Radio link	Bin to farmhouse over 1 km or more with one farm building in the path; 95 % or more of packets delivered	Link budget; later site survey
R8	Power autonomy	5 days or more with no solar input at -20 °C with the fan running continuously; recovers from 50 % charge in 3 winter days	Power budget
R9	Environment	Operate at -30 °C to +50 °C; controller IP66; pods dust-tight (IP6X) with a vapor-permeable membrane; survive one phosphine fumigation per season	Datasheets and design review; later exposure test
R10	Mechanical strength	Rope minimum breaking load 4 times or more the design pull-down force of 2.5 kN (estimate); roof hanger rated by the bin maker for the design force	Force estimate; bin maker's data
R11	Install and service without grain entry	Installed with the bin empty, from the roof manhole and floor under a confined-space and fall-protection procedure; no GrainGuard task needs anyone in a bin that holds grain	Installation sequence review
R12	Electrical isolation	Controller, cable and pods at 15 V DC or less; the relay kit is the only mains-connected item, installed by a licensed electrician, with 2.5 kV or more isolation between control input and contacts	Circuit review; relay datasheet
R13	Affordable	\$250 or less in parts per bin	Priced BOM (bom/ bom.csv)

ID	Requirement	Target	Verification (TRL 3 or later)
R14	Local data	All readings and fan hours stored on the farm; no cloud account needed	Architecture review

Table 2. Status of the concept against each requirement (estimates, see GGD-PRC-001).

ID	Status at TRL 2	Basis
R1	Met on paper	Six SHT45 pods at 1.0 m pitch; ± 0.1 °C typical sensor accuracy
R2	Met on paper for 20 % to 75 % RH; not met above about 80 % RH	Published EMC prediction error of 0.25 to 0.65 points (dry basis) with ± 2 % RH sensors; error grows fast above 70 % to 80 % RH, that is for wet grain
R3	Met on paper	Sensor accuracy; shield error unverified
R4	Met by design	Rule in GGD-PRC-001
R5	Met by design	Normally open relay and hand-off-auto switch
R6	At risk	Met for the center core only; heating more than about 0.5 m from the cable is not detected
R7	Met on paper	About 48 dB of margin over free space at 2 km (estimate)
R8	Not met	About 4.8 days with the fan running continuously; about 7.6 days at 8 h of fan time per day
R9	At risk	Phosphine corrodes copper; pod and connector protection unverified
R10	Rope met; roof capacity not met until confirmed	6 mm rope about 20 kN, 8 times the design force; older bin roofs may not have a rated hanger
R11	Met by design	Installation sequence in GGD-PRC-001
R12	Met by design	Solar 12 V controller; DIN relay kit
R13	Not met	About \$271 per bin, about \$293 with the shared farmhouse receiver
R14	Met by design	Farmhouse receiver stores data locally

■ Assumptions

- Grain properties: shelled corn at 721 kg/m³ (25.4 kg per 0.0352 m³ bushel) and a specific heat of about 1.9 kJ/(kg·K).
- EMC from the ASABE D245.6 modified Henderson equation for corn; accuracy of RH-based moisture prediction as reported by [USDA ARS](#).
- The fan warms the air about 1 °C (estimate); to be measured at TRL 3.
- Cooling front time of 75 to 120 h at 0.2 cfm/bu, from extension guidance and a sensible-heat balance.
- Design pull-down force of 2.5 kN for a thin 6 mm cable is an estimate. Forces up to 4.7 kN were measured on the largest commercial cable during unloading ([Illinois Experts](#), [ASABE](#)); thinner cables see less force, but this is unverified.